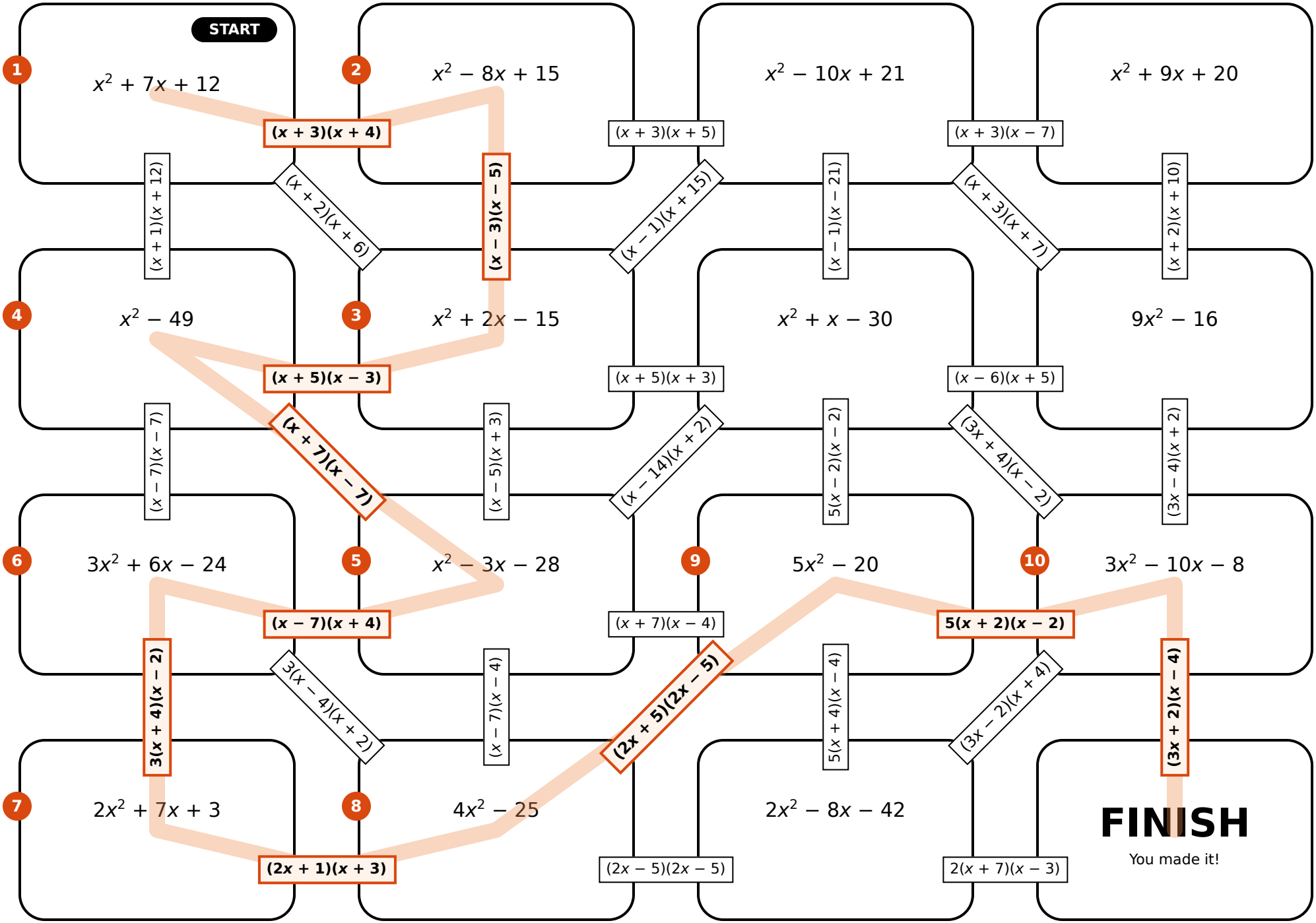
Factoring Maze ANSWER KEY

Start in the top-left room. Factor completely, find the door with your answer, and go through it. Show your work on your answer grid.



Factoring Maze: Answer Grid **ANSWER KEY**

Write each answer in the order you found it, and show your work for that room.

1	Answer: **$(x + 3)(x + 4)$**	$3 \cdot 4 = 12 \;\rightarrow\; 3 + 4 = 7$
2	Answer: **$(x - 3)(x - 5)$**	$(-3)(-5) = 15 \;\rightarrow\; -3 + (-5) = -8$
3	Answer: **$(x + 5)(x - 3)$**	$5(-3) = -15 \;\rightarrow\; 5 + (-3) = 2$
4	Answer: **$(x + 7)(x - 7)$**	$x^2 - 7^2 \;\rightarrow\; (x + 7)(x - 7)$
5	Answer: **$(x - 7)(x + 4)$**	$(-7)(4) = -28 \;\rightarrow\; -7 + 4 = -3$
6	Answer: **$3(x + 4)(x - 2)$**	$3(x^2 + 2x - 8) \;\rightarrow\; 3(x + 4)(x - 2)$
7	Answer: **$(2x + 1)(x + 3)$**	$ac = 6$: $1 \cdot 6$, $1 + 6 = 7 \;\rightarrow\; 2x^2 + x + 6x + 3 \;\rightarrow\; (2x + 1)(x + 3)$
8	Answer: **$(2x + 5)(2x - 5)$**	$(2x)^2 - 5^2 \;\rightarrow\; (2x + 5)(2x - 5)$
9	Answer: **$5(x + 2)(x - 2)$**	$5(x^2 - 4) \;\rightarrow\; 5(x + 2)(x - 2)$
10	Answer: **$(3x + 2)(x - 4)$**	$ac = -24$: $2 \cdot (-12)$, $2 - 12 = -10 \;\rightarrow\; 3x^2 + 2x - 12x - 8 \;\rightarrow\; (3x + 2)(x - 4)$